

FedGuard Governance Control Plane & Conclave System Architecture

1. Participant Layer: Multi-Institutional Edge Nodes & Hardware Root-of-Trust

Client Node u (Hospital / Bank)

- mTLS 1.3 X.509 Certificate
- TPM 2.0 PCR Attestation Quote
- Local PyTorch Model Update Δ_u

mTLS / Quote

Client Node v (Enterprise Silo)

- mTLS 1.3 X.509 Certificate
- TPM 2.0 PCR Attestation Quote
- Local PyTorch Model Update Δ_v

mTLS / Quote

Client Node N (Edge Participant)

- mTLS 1.3 X.509 Certificate
- Software Certificate Fallback
- Local PyTorch Model Update Δ_N

mTLS / Fallback

2. FedGuard Decoupled Governance Control Plane Engine

ABAC / RBAC Policy Engine

- X.509 Certificate Validation
- Role & Attribute Matching
- TPM 2.0 Quote Measurement
- Dynamic Access Enforcement
- **HIPAA / GDPR Mappings**

Policy Authorization

Mode Handler & SecAgg

- **Mode Selection:**
 - SECURE_AGGREGATION
 - BYZANTINE_RESILIENT
- Pairwise ECDH Key Exchange
- Shamir Secret Sharing (k-of-N)

SecAgg / Mask Rules

Privacy & Unlearning Engine

- Rényi DP Budget Accountant (Tracks (ϵ_t, δ_t) vs $\epsilon_{\max}$)
- Gaussian Noise Injector
- GDPR Art. 17 Revocation Handler
- **FedEraser Unlearn Trigger**

DP Noise / Revocation

3. Cryptographic Execution & Robust Aggregation Layer (Flower Integration)

Modular Tensor Quantizer

- Scale $\gamma = 2^{16}$, Modulus $R = 2^{32-1}$
- $y_u = x_u + \sum PRG(s) \pmod R$

Byzantine Robust Aggregators

- Trimmed Mean ($\beta=0.2$), Median
- Krum Coordinate Distance Filter

Global Model Orchestration

- Aggregated Update Δ_{global}
- Flower FL Engine Dispatch

Audit Event Logging

Purify Checkpoints

4. Immutable Storage & Verification Layer

SHA-256 Hash-Chained Audit Ledger

$H_k = \text{SHA-256}(H_{(k-1)} \parallel \text{Event}_k \parallel T_{\text{UTC}} \parallel \text{Sign}_{\text{server}})$

Tamper-Evident Historical Audit Event Verification

GDPR Art. 17 Machine Unlearning Store

FedEraser Gradient History Un-rolling & Purification

Automated Model Parameter Reconstruction on Eviction